

# ELL8299 Assignment Report

## Part 1

### Training

First, I ran the recommended base line configuration, as mentioned in the assignment instructions.

To be specific, the configuration was

Context\_length: 64

Num\_layers: 3

Num\_attention\_heads : 8

Hidden\_dim: 300

The train and validation loss curves are shown below.

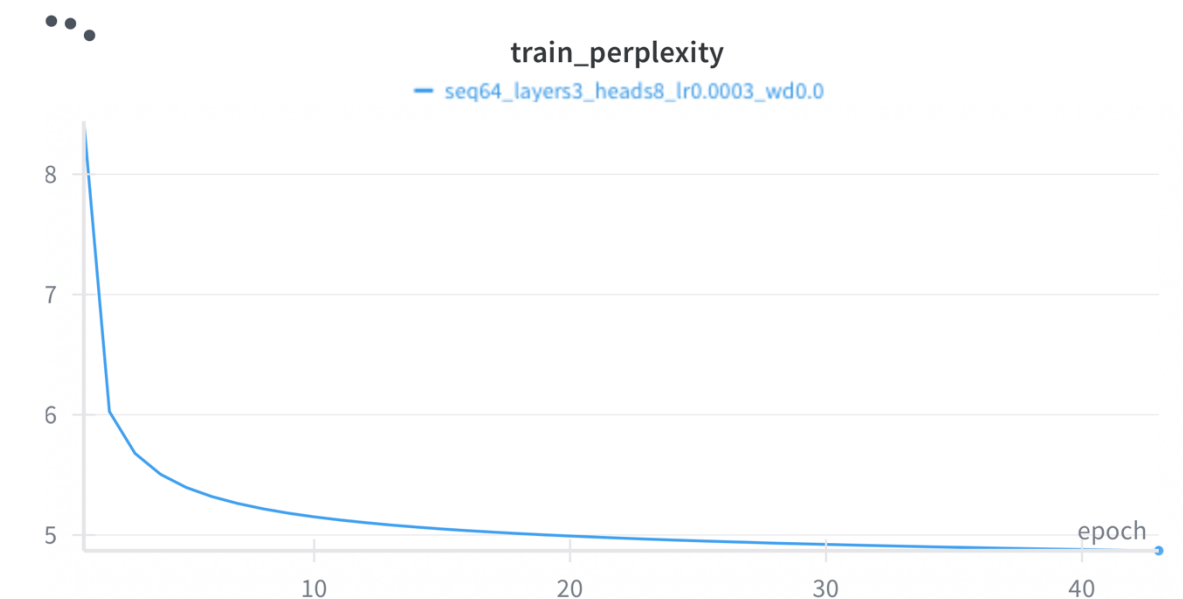


*1: Train/Val loss curve for baseline config*

The final epoch train loss was: **1.58296**

The final epoch val loss was: **1.52328**

The train perplexity curve for the baseline model looks as follows:



2Train Perplexity over epochs for baseline

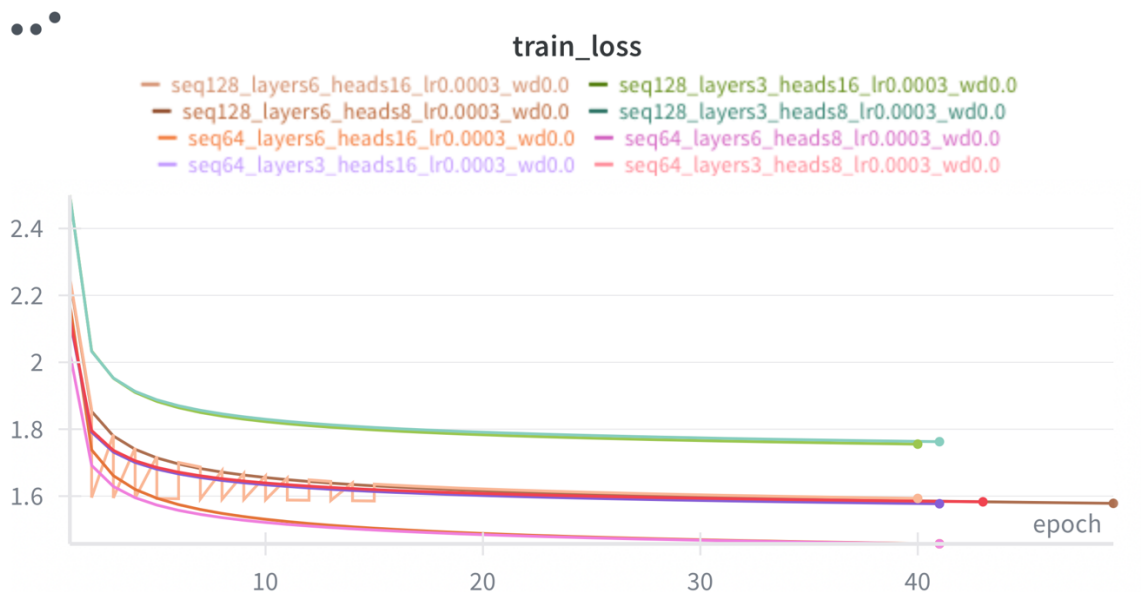
The final epoch train perplexity was: 4.86933

To experiment with multiple configurations, we ran a grid test over the following parameter choices:

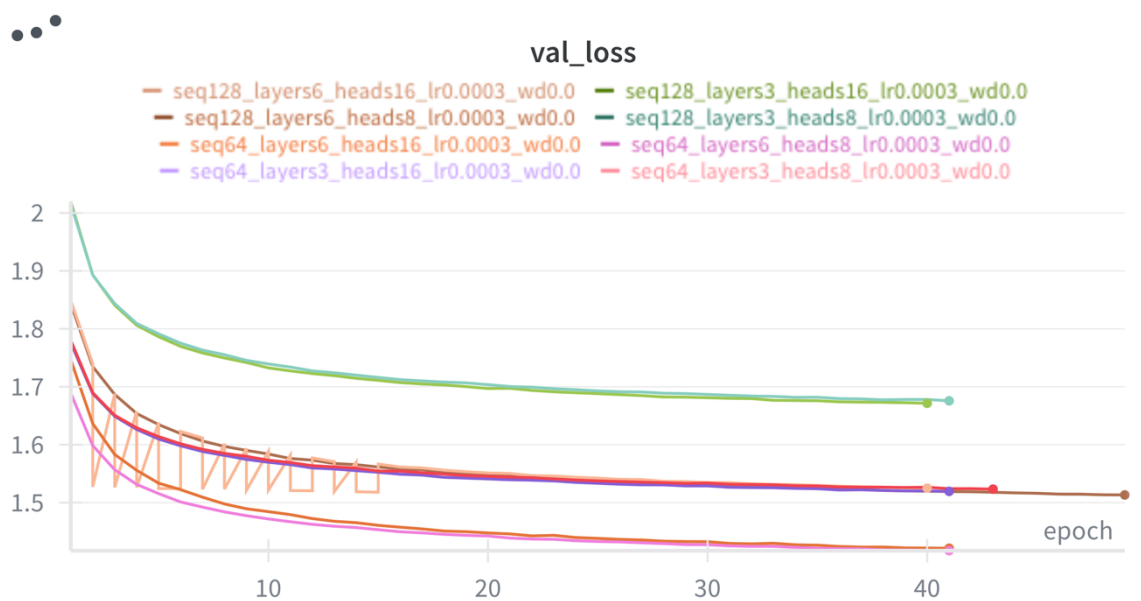
```
sequence_lengths=(64, 128)
num_layers=(3, 6)
num_heads=(8 16)
```

which resulted in a total of 8 unique configurations.

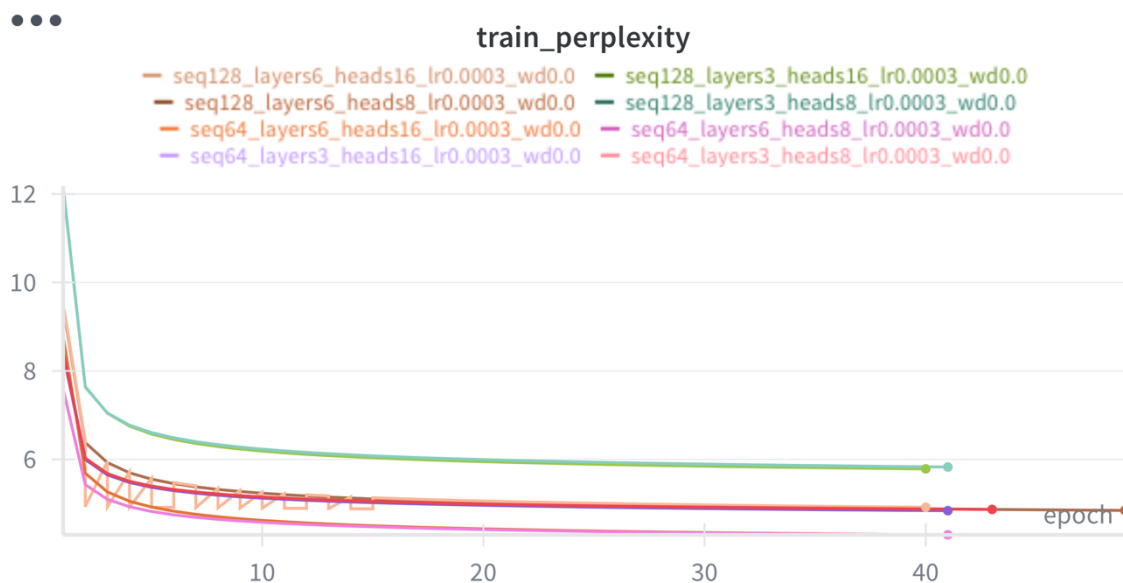
Below are the train and val loss plots for all the configurations



3 Train Loss over epochs for all configs



4 Val Loss over epochs for all configs



5 Train Perplexity over epochs for all configs

Upon examining these results, the model config with best metrics on the test sets was

Context\_length: 64, layers:3, heads: 8

| Metric          | Metric Value |
|-----------------|--------------|
| Test Loss       | 1.42533      |
| Test BLEU       | 0.37189      |
| Test Perplexity | 4.15922      |

### Inference

We select the model with above configuration to perform inference experiments.

Average Perplexity per token on samples from TinyStories validation set: **1.56134**

Average BLEU score on 50 samples from TinyStories validation set: **0.0133670**

Average time taken per token with Sampling/Greedy Decoding: **0.00621**

Below, some of the prompt/generated outputs and the relevant stats with them are showcased

**Prompt:** once upon a time ,

**Generated Text:** there was a little girl named lily . she loved to play outside in the sunshine . one day , she saw a big , scary dog . the dog barked loudly and lily was scared . her mommy told her not to worry and that the dog was friendly . lily went to her mommy and said

**Perplexity per token:** 1.222

**BLEU Score:** 0.04932

**Avg time per token:** 0.00995 seconds

**Prompt:** tom and mia were going

**Generated Text:** to the park with mom . they liked the park . it had swings , slides and a big sandbox . they put on their hats and coats and went to the park . they saw a big slide . it was very high and fast . they wanted to go on the slide . " mom ,

**Perplexity per token:** 1.1894

**BLEU Score:** 0.0010

**Avg time per token:** 0.0064 seconds

**Prompt:** one day , a small

**Generated Text:** boy named tim went to the park with his mom . they saw a big tree with a swing . tim wanted to play on the swing , but his mom said , " no , tim . you ca n't swing on the swing . " tim felt sad . he wanted to play on the swing

**Perplexity per token:** 1.2566

**BLEU Score:** 0.0

**Avg time per token:** 0.00605 seconds

The low BLEU scores can be explained by the high degree of difference between the ground truth text and the model generated completion.

Attention heads:

## **Part 2**

### Beam Search

| k (Beam width) | Mean PPL / token | Mean BLEU | Mean Time per Token (seconds) |
|----------------|------------------|-----------|-------------------------------|
| 5              | 1.584            | 0.0223    | 0.012                         |
| 10             | 1.606            | 0.012     | 0.0227                        |

**Prompt:** jake loved to jog .

**Generated Text:** every morning , he would put on his shoes and go outside . he would put on his shoes and jog around the park . one day , jake was jogging in the park when he saw a big tree . he stopped to take a closer look . it was a big tree with lots of leaves

**Perplexity per token:** 1.9379007337606062

**BLEU Score:** 0.039369751304112174

**Avg time per token:** 0.014949527280084017

**Prompt:** once upon a time ,

**Generated Text:** there was a little girl named lily . she loved to play outside and explore the world around her . one day , she went on a walk with her mommy and daddy . they walked and walked until they found a big tree with a hole in it . lily asked her mommy , " what 's

**Perplexity per token:** 1.4346947552748277

**BLEU Score:** 0.0

**Avg time per token:** 0.012088307018937736

**Prompt:** once upon a time there

**Generated Text:** was a little girl named lucy . she was three years old and loved to explore the world around her . one day , lucy decided to go for a walk in the park . as she was walking , she saw a big tree . she wanted to climb it , so she started to climb .

**Perplexity per token:** 1.6814239292078796

**BLEU Score:** 0.006125720212951681

**Avg time per token:** 0.012189692464368096

**Prompt:** once upon a time ,

**Generated Text:** there was a little girl named lily . she loved to play outside and explore the world around her . one day , she went on a walk with her mommy and daddy . they walked and walked until they found a big tree with a hole in it . lily asked her mommy , " what 's

**Perplexity per token:** 1.4346 seconds

**BLEU Score:** 0.0552 seconds

**Avg time per token:** 0.0120 seconds

**Prompt:** once upon a time ,

**Generated Text:** there was a little girl named lily . she loved to play outside and explore the world around her . one day , she went on a walk with her mommy and daddy . they walked and walked until they found a big tree with a hole in it . lily asked her mommy , " what 's

**Perplexity per token:** 1.4346 seconds

**BLEU Score:** 0.0109 seconds

**Avg time per token:** 0.0121 seconds